

AI/ML Internship – Day 26 Notes & Tasks

Module 5: Pandas Introduction (Data Analysis Library)

Part 1: What is Pandas?

Definition

 Pandas is a powerful Python library used for:

- Data analysis
- Data manipulation
- Data cleaning

 It is built on top of NumPy.

Part 2: Why Pandas is Important?

 In AI/ML:

- Real-world data is messy
- Pandas helps organize data easily

 Used for:

- Reading CSV files
 - Cleaning datasets
 - Handling missing values
 - Data analysis
-

Part 3: Theory Answers (IMPORTANT)

1. What is Pandas?

 Pandas is a Python library used for working with structured data.

2. Why Pandas is Used?

👉 Pandas helps:

- Analyze data
 - Clean data
 - Process datasets quickly
-

◆ 3. What is a DataFrame?

👉 DataFrame is a table-like structure in Pandas.

👉 It contains:

- Rows
- Columns

Similar to Excel table.

◆ 4. What is a Series?

👉 Series is a one-dimensional data structure in Pandas.

👉 Similar to NumPy 1D array.

◆ 5. Difference Between NumPy and Pandas

NumPy	Pandas
Numerical operations	Data analysis
Arrays	Tables/DataFrames
Faster math operations	Easier data handling

📌 Part 4: Installing Pandas

`pip install pandas`

📌 Part 5: Import Pandas

`import pandas as pd`

🚀 Part 6: Creating a Series

```
import pandas as pd

data = pd.Series([10, 20, 30, 40])

print(data)
```

◆ Custom Index

```
data = pd.Series([85, 90, 78], index=["Math", "Science", "English"])

print(data)
```

🚀 Part 7: Creating DataFrame

◆ Using Dictionary

```
data = {
    "Name": ["Rahul", "Anu", "Arjun"],
    "Marks": [85, 90, 78]
}

df = pd.DataFrame(data)

print(df)
```

🚀 Part 8: Understanding DataFrame

👉 DataFrame contains:

- Rows
 - Columns
 - Index
-

◆ Access Columns

```
print(df["Name"])
```

◆ Access Multiple Columns

```
print(df[["Name", "Marks"]])
```

✦ Part 9: DataFrame Information

◆ Head Function

```
print(df.head())
```

👉 Shows first 5 rows.

◆ Tail Function

```
print(df.tail())
```

👉 Shows last rows.

◆ Info Function

```
print(df.info())
```

👉 Shows:

- Columns
 - Data types
 - Null values
-

✦ Part 10: Real-World Thinking

👉 In AI projects:

- Datasets are mostly CSV/Excel files
- Pandas is used before Machine Learning

👉 Example:

- Student dataset
- Sales dataset

- Hospital records
 - Weather data
-

Part 11: Visualization of DataFrame

```
+-----+-----+
| Name  | Marks |
+-----+-----+
| Rahul | 85    |
| Anu   | 90    |
| Arjun | 78    |
+-----+-----+
```

Tasks for Day 26

Task 1: Theory (Write Answers)

1. What is Pandas?
 2. Why Pandas is important?
 3. What is Series?
 4. What is DataFrame?
 5. Difference between NumPy and Pandas
-

Task 2: Series Practice

1. Create Series
 2. Create Series with custom index
 3. Access Series values
-

Task 3: DataFrame Practice

1. Create DataFrame using dictionary
2. Print DataFrame

3. Add one more column
-

Task 4: Data Access

1. Access single column
 2. Access multiple columns
 3. Print specific rows
-

Task 5: Data Information

1. Use head()
2. Use tail()
3. Use info()